

Exogeneity

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EXOGENEITY¹

BY ROBERT F. ENGLE, DAVID F. HENDRY, AND JEAN-FRANCOIS RICHARD

Definitions are proposed for *weak* and *strong* exogeneity in terms of the distribution of *observable* variables. The objectives of the paper are to clarify the concepts involved, isolate the essential requirements for a variable to be exogenous, and relate them to notions of predeterminedness, *strict* exogeneity and causality in order to facilitate econometric modelling. Worlds of parameter change are considered and exogeneity is related to structural invariance leading to a definition of *super* exogeneity. Throughout the paper, illustrative models are used to exposit the analysis.

1. INTRODUCTION

SINCE "EXOGENEITY" IS FUNDAMENTAL to most empirical econometric modelling, its conceptualization, its role in inference, and the testing of its validity have been the subject of extensive discussion (see inter alia, Koopmans [21], Orcutt [28], Marschak [26], Phillips [29], Sims [38, 39], Geweke [13, 14] and Richard [32]). Nevertheless, as perusal of the literature (and especially econometrics textbooks) quickly reveals, precise definitions of "exogeneity" are elusive and consequently, it is unclear exactly what is entailed for *inference* by the discovery that a certain variable is "exogenous" on any given definition. Moreover, the motivation underlying various "exogeneity" concepts has not always been stated explicitly so that their relationships to alternative notions of "causality" (see Wiener [42], Strotz and Wold [40], Granger [16], and Zellner [45]) remain ambiguous. This results in part because some definitions have been formulated for limited classes of models so that appropriate generalizations such as to nonlinear or non-Gaussian situations are not straightforward, while others are formulated in terms involving unobservable disturbances from relationships which contain unknown parameters. Whether or not such disturbances satisfy orthogonality conditions with certain observables may be a matter of construction or may be a testable hypothesis and a clear distinction between these situations is essential.

In this paper, definitions are proposed for *weak* and *strong* exogeneity in terms of the distributions of observable variables,² thereby explicitly relating these

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²The emphasis on observables does not preclude formulating theories in terms of unobservables (e.g., "permanent" components, expectations, disturbances, etc.), but these should be integrated out first in order to obtain an operational model to which our concepts may be applied.

concepts to the likelihood function and hence efficient estimation:³ essentially, a variable z_t in a model is defined to be weakly exogenous for estimating a set of parameters λ if inference on λ conditional on z_t involves no loss of information. Heuristically, given that the joint density of random variables (y_t, z_t) always can be written as the product of y_t conditional on z_t times the marginal of z_t , the weak exogeneity of z_t entails that the precise specification of the latter density is irrelevant to the analysis, and, in particular that all parameters which appear in this marginal density are nuisance parameters. Such an approach builds on the important paper by Koopmans [21] using recently developed concepts of statistical inference (see e.g., Barndorff-Nielsen [1], and Florens and Mouchart [10]). If in addition to being weakly exogenous, z_t is *not caused* in the sense of Granger [16] by any of the endogenous variables in the system, then z_t is defined to be strongly exogenous.

The concept of exogeneity is then extended to the class of models where the mechanism generating z_t changes. Such changes could come about for a variety of reasons; one of the most interesting is the attempt by one agent to control the behavior of another. If all the parameters λ of the conditional model are invariant to any change in the marginal density of z_t , and z_t is weakly exogenous for λ , then z_t is said to be *super* exogenous. That is, changes in the values of z_t or its generating function will not affect the conditional relation between y_t and z_t . This aspect builds on the work of Frisch [12], Marschak [26], Hurwicz [20], Sims [39], and Richard [32].

The paper is organized as follows: formal definitions of weak, strong, and super exogeneity are introduced in Section 2; and, to ensure an unambiguous discussion, the familiar notions of predeterminedness, strict exogeneity, and Granger noncausality are also defined. These are then discussed in the light of several examples in Section 3. The examples illustrate the relations between the concepts in familiar models showing the importance of each part of the new definitions and showing the incompleteness of the more conventional notions. Special attention is paid to the impact of serial correlation. The analysis is then applied to potentially incomplete dynamic simultaneous equations systems in Section 4. The conclusion restates the main themes and implications of the paper.

1.1 Notation

Let $x_t \in R^n$ be a vector of *observable* random variables generated at time t , on which observations ($t = 1, \dots, T$) are available. Let X_t^1 denote the $t \times n$ matrix:

$$(1) \quad X_t^1 = (x_1, \dots, x_t)'$$

and let X_0 represent the (possibly infinite) matrix of initial conditions. The analysis is conducted conditionally on X_0 . For a discussion of marginalization

³Throughout the paper, the term "efficient estimation" is used as a shorthand for "conducting inference without loss of relevant information," and does not entail any claims as to e.g., the efficiency of particular estimators in small samples.

with respect to initial conditions, see Engle, Hendry, and Richard [8], hereafter EHR. The information available at time t is given by

$$(2) \quad X_{t-1} = \begin{bmatrix} X_0 \\ X_{t-1}^1 \end{bmatrix}.$$

The process generating the T observations is assumed to be continuous with respect to some appropriate measure and is represented by the joint data density function $D(X_T^1 | X_0, \theta)$ where θ , in the interior of Θ , is an (identified) vector of unknown parameters. The likelihood function of θ , given the initial conditions X_0 , is denoted by $L^0(\theta; X_T^1)$.

Below, $f_N^n(\cdot | \mu, \Sigma)$ denotes the n -dimensional normal density function with mean vector μ and covariance matrix Σ . The notation $x_t \sim IN(\mu, \Sigma)$ reads as “the vectors $x_1, \dots, x_T$ are identically independently normally distributed with common mean vector μ and covariance matrix Σ .” C_n denotes the set of symmetric positive definite matrices.

The vector x_t is partitioned into

$$(3) \quad x_t = \begin{bmatrix} y_t \\ z_t \end{bmatrix}, \quad y_t \in R^p, \quad z_t \in R^q, \quad p + q = n.$$

The matrices X_0 , X_t^1 , and X_t are partitioned conformably:

$$(4) \quad X_0 = (Y_0 Z_0), \quad X_t^1 = (Y_t^1 Z_t^1), \quad X_t = (Y_t Z_t).$$

The expressions “ $x_t \parallel y_t$ ” and “ $x_t \parallel y_t | w_t$ ” read respectively as “ x_t and y_t are independent (in probability)” and “conditionally on w_t , x_t and y_t are independent.” In our framework it is implicit that all such independence statements are *conditional* on θ . The operator $\sum$ denotes a summation which starts at $i = 1$ and is over all relevant lags.

2. DEFINITIONS

Often the objective of empirical econometrics is to model how the observation x_t is generated conditionally on the past, so we factorize the joint data density as

$$(5) \quad D(X_T^1 | X_0, \theta) = \prod_{t=1}^T D(x_t | X_{t-1}, \theta)$$

and focus attention on the conditional density functions $D(x_t | X_{t-1}, \theta)$. These are assumed to have a common functional form with a *finite*⁴ dimensional parameter space Θ .

The following formal definitions must be introduced immediately to ensure an unambiguous discussion, but the examples presented below attempt to elucidate their content; the reader wishing a general view of the paper could proceed fairly rapidly to Section 3 and return to this section later.

⁴It is assumed that the dimensionality of Θ is sufficiently small relative to nT that it makes sense to discuss, e.g., “efficient” estimation.

2.1. Granger Noncausality

For the class of models defined by (5), conditioned throughout on X_0 , Granger [16] provides a definition of noncausality which can be restated as:

DEFINITION 2.1: Y_{t-1}^1 does not Granger cause z_t with respect to X_{t-1} if and only if

$$(6) \quad D(z_t | X_{t-1}, \theta) = D(z_t | Z_{t-1}, Y_0, \theta),$$

i.e., if and only if

$$(7) \quad z_t \parallel Y_{t-1}^1 | Z_{t-1}, Y_0.$$

If condition (6) holds over the sample period, then the joint data density $D(X_T^1 | X_0, \theta)$ factorizes as

$$(8) \quad D(X_T^1 | X_0, \theta) = \left[\prod_{t=1}^T D(y_t | z_t, X_{t-1}, \theta) \right] \left[\prod_{t=1}^T D(z_t | Z_{t-1}, Y_0, \theta) \right]$$

where the last term is $D(Z_T^1 | X_0, \theta)$ and the middle term is therefore $D(Y_T^1 | Z_T^1, X_0, \theta)$.

Where no ambiguity is likely, condition (6) is stated below as “ y does not Granger cause z .” Note that the definition in Chamberlain [3] is the same as 2.1.

2.2. Predeterminedness and Strict Exogeneity

Consider a set of $g \leq n$ behavioral relationships (whose exact interpretation is discussed in Section 4 below):

$$(9) \quad B^* x_t + \sum C^*(i) x_{t-i} = u_t$$

where B^* and $\{C^*(i)\}$ are $g \times n$ matrix functions of θ , with rank $B^* = g$ almost everywhere in Θ , and u_t is the corresponding “disturbance.”

The following definitions are adapted from Koopmans and Hood [22]—see also Christ [4, Chs. IV.4, VI.4] and Sims [39].

DEFINITION 2.2: z_t is *predetermined* in (9) if and only if

$$(10) \quad z_t \parallel u_{t+i} \quad \text{for all } i \geq 0.$$

DEFINITION 2.3: z_t is *strictly exogenous*⁵ in (9) if and only if

$$(11) \quad z_t \parallel u_{t+i} \quad \text{for all } i.$$

⁵We use the term “strictly exogenous” where some authors use “exogenous” to distinguish this concept from that introduced below.

The connections between strict exogeneity and Granger noncausality have been discussed by several authors—and in particular by Sims [39] and Geweke [13]—for complete dynamic simultaneous equations models. This issue is reconsidered in Section 4. See also the discussion in Chamberlain [3] and Florens and Mouchart [11].

2.3. *Parameters of Interest*

Often a model user is not interested in all the parameters in θ , so that his (implicit) loss function depends only on some functions of θ , say:

$$(12) \quad f: \Theta \rightarrow \Psi; \quad \theta \rightarrow \psi = f(\theta).$$

These functions are called *parameters of interest*. Parameters may be of interest, e.g., because they are directly related to theories the model user wishes to test concerning the structure of the economy. Equally, in seeking empirical econometric relationships which are constant over the sample period and hopefully over the forecast period, parameters which are *structurally invariant* (see Section 2.6), are typically of interest.

Since models can be parameterized in infinitely many ways, parameters of interest need not coincide with those which are chosen to characterize the data density (e.g., the mean vector and the covariance matrix in a normal framework). Consider, therefore, an arbitrary one-to-one transformation or reparameterization:

$$(13) \quad h: \Theta \rightarrow \Lambda; \quad \theta \rightarrow \lambda = h(\theta)$$

together with a partition of λ into (λ_1, λ_2) . Let Λ_i denote the set of admissible values of λ_i . The question of whether or not the parameters of interest are *functions* of λ_1 plays an essential role in our analysis: that is, whether there exists a function ϕ ,

$$(14) \quad \phi: \Lambda_1 \rightarrow \Psi; \quad \lambda_1 \rightarrow \psi = \phi(\lambda_1)$$

such that

$$(15) \quad \text{for all } \lambda \in \Lambda, \quad \psi = f[h^{-1}(\lambda)] \equiv \phi(\lambda_1).$$

When (15) holds, λ_2 is often called a *nuisance parameter*.⁶

2.4. *Sequential Cuts*

Let $x_i \in R^n$ be partitioned as in (3) and let $\lambda = (\lambda_1, \lambda_2)$ be a reparameterization as in (13). The following definition is adapted from Florens and Mouchart [10]

⁶The concept of *nuisance* parameter is, however, ambiguous. Whether or not a parameter is a *nuisance* parameter critically depends on which (re)parameterization is used. If, for example, $\theta = (\alpha, \beta)$ and β is the sole parameter of interest, then α is a nuisance parameter. In contrast, a reparameterization using (α, γ) where $\gamma = \beta/\alpha$ entails that β is not a function of γ alone, and so α is not a *nuisance* parameter.

who generalized the notion of cut discussed (e.g.) by Barndorff-Nielsen [1] to dynamic models:

DEFINITION 2.4: $[(y_t|z_t; \lambda_1), (z_t; \lambda_2)]$ operates a (classical) *sequential cut* on $D(x_t|X_{t-1}, \lambda)$ if and only if

$$(16) \quad D(x_t|X_{t-1}, \lambda) = D(y_t|z_t, X_{t-1}, \lambda_1)D(z_t|X_{t-1}, \lambda_2)$$

where λ_1 and λ_2 are *variation free*, i.e.,

$$(17) \quad (\lambda_1, \lambda_2) \in \Lambda_1 \times \Lambda_2.$$

Since Λ_i denotes the set of admissible values of λ_i , condition (17) requires in effect that λ_1 and λ_2 should not be subject to “cross-restrictions,” whether exact or inequality restrictions, since then the range of admissible values for λ_i would vary with λ_j ($i, j = 1, 2; j \neq i$).

2.5. Weak and Strong Exogeneity

The following definitions are adapted from Richard [32]. As in (12), ψ denotes the parameter of interest.

DEFINITION 2.5: z_t is *weakly exogenous* over the sample period for ψ if and only if there exists a reparameterization with $\lambda = (\lambda_1, \lambda_2)$ such that

- (i) ψ is a function of λ_1 (as in (15)),
- (ii) $[(y_t|z_t; \lambda_1), (z_t; \lambda_2)]$ operates a sequential cut.

DEFINITION 2.6: z_t is *strongly exogenous* over the sample period for ψ if and only if it is weakly exogenous for ψ and in addition

- (iii) y does not Granger cause z .

When (ii) holds, $L^0(\lambda; X_T^1)$ factorizes as in

$$(18) \quad L^0(\lambda; X_T^1) = L_1^0(\lambda_1; X_T^1) L_2^0(\lambda_2; X_T^1),$$

where

$$(19) \quad L_1^0(\lambda_1; X_T^1) = \prod_{t=1}^T D(y_t|z_t, X_{t-1}, \lambda_1),$$

$$(20) \quad L_2^0(\lambda_2; X_T^1) = \prod_{t=1}^T D(z_t|X_{t-1}, \lambda_2),$$

and the two factors in (18) can be analyzed independently of each other (which, irrespective of whether or not (i) holds may considerably reduce the computational burden). If in addition (i) holds, then all the sample information concerning the parameter of interest ψ can be obtained from the partial likelihood function $L_1^0(\lambda_1; X_T^1)$. If it were known (or assumed a priori) that z_t was weakly exogenous for ψ , then the marginal process $D(z_t | X_{t-1}, \lambda_2)$ would not even need to be specified. However, tests of the weak exogeneity of z_t for ψ , as described in Section 6.1 of EHR and Engle [6], evidently require that the *joint* model $D(x_t | X_{t-1}, \lambda)$ be specified.

The factorization (18)–(20) does not entail that the conditional process generating $\{y_t | z_t\}$ and the marginal process generating $\{z_t\}$ can be separated from each other, i.e., for example, that z_t can be treated as “fixed” in the conditional model $D(y_t | z_t, X_{t-1}, \lambda_1)$ since lagged values of y_t may still affect the process generating z_t .⁷ Factorizing the joint data density $D(X_T^1 | X_0, \lambda)$ requires an additional assumption and this is precisely the object of Granger noncausality. When both (ii) and (iii) hold we can factorize $D(X_T^1 | X_0, \lambda)$ as in

$$(21) \quad D(X_T^1 | X_0, \lambda) = D(Y_T^1 | Z_T^1, X_0, \lambda_1) D(Z_T^1 | X_0, \lambda_2),$$

where

$$(22) \quad D(Y_T^1 | Z_T^1, X_0, \lambda_1) = \prod_{t=1}^T D(y_t | z_t, X_{t-1}, \lambda_1),$$

$$(23) \quad D(Z_T^1 | X_0, \lambda_2) = \prod_{t=1}^T D(z_t | Z_{t-1}, Y_0, \lambda_2).$$

It must be stressed that the definition of Granger noncausality as given in (6) and (8), includes *no* assumption about the *parameters*. This is precisely why it must be completed by an assumption of weak exogeneity in order to entail a complete separation of the processes generating respectively $\{y_t | z_t\}$ and $\{z_t\}$.

2.6. Structural Invariance and Super Exogeneity

A closely related issue of statistical inference is parameter constancy. Over time, it is possible that some of the parameters of the joint distribution may change perhaps through changing tastes, technology, or institutions such as government policy making. For some classes of parameter change or “interventions” there may be parameters which remain constant and which can be estimated without difficulty even though interventions occur over the sample period. This is a familiar assumption about parameters in econometrics which is here called invariance. Just as weak exogeneity sustains conditional inference

⁷It follows that, unless y does not Granger cause z , $L_1^0(\lambda_1; X_T^1)$ is not *sensu stricto* a likelihood function, although it is often implicitly treated as such in the econometric literature, but it is a valid basis for inferences about ψ , *provided* z_t is *weakly exogenous* for ψ .

within a regime, we develop the relevant exogeneity concept for models subject to a particular class of regime changes.

DEFINITION 2.7: A parameter is *invariant* for a class of interventions if it remains constant under these interventions. A model is invariant for such interventions if all its parameters are.

DEFINITION 2.8: A conditional model is *structurally invariant* if all its parameters are invariant for any change in the distribution of the conditioning variables.⁸

Since weak exogeneity guarantees that the parameters of the conditional model and those of the marginal model are variation free, it offers a natural framework for analyzing the structural invariance of parameters of conditional models. However, by itself, weak exogeneity is neither necessary nor sufficient for structural invariance of a conditional model. Note, first, that the conditional model may be structurally invariant without its parameters providing an estimate of the parameters of interest. Conversely, weak exogeneity of the conditioning variables does not rule out the possibility that economic agents change their behavior in relation to interventions. That is, even though the parameters of interest and the nuisance parameters are variation free over any given regime, where a regime is characterized by a fixed distribution of the conditioning variables, their variations between regimes may be related. This will become clear in the examples.

The concept of structurally invariant conditional models characterizes the conditions which guarantee the appropriateness of “policy simulations” or other control exercises, since any change in the distribution of the conditioning variables has no effect on the conditional submodel and therefore on the conditional forecasts of the endogenous variables. This requirement is clearly very strong and its untested assumption has been criticized in conventional practice by Lucas [23] and Sargent [35].

To sustain conditional inference in processes subject to interventions, we define the concept of *super exogeneity*.

DEFINITION 2.9: z_t is *super exogenous* for ψ if z_t is weakly exogenous for ψ and the conditional model $D(y_t | z_t, X_{t-1}, \lambda_1)$ is structurally invariant.

Note that Definition 2.9 relates explicitly to *conditional* submodels: since estimable models with invariant parameters but no weakly exogenous variables are easily formulated (see Example 3.2 below), super exogeneity is a sufficient but not a necessary condition for valid inference under interventions (see e.g. the discussion of feasible policy analyses under rational expectations in Wallis [41] and the formulation in Sargent [35]).

⁸The definition can always be restricted to a specific class of distribution changes. This will implicitly be the case in the examples which are discussed in Section 3.

It is clear that any assertion concerning super exogeneity is refutable in the data for past changes in $D(z_t | X_{t-1}, \lambda_2)$ by examining the behavior of the conditional model for invariance when the parameters of the exogenous process changed. For an example of this see Hendry [18]. However, super exogeneity for all changes in the distribution of z_t must remain a conjecture until refuted, both because nothing precludes agents from simply changing their behavior at a certain instant and because only a limited range of interventions will have occurred in any given sample period (compare the notion of nonexcitation in Salmon and Wallis [34]). Such an approach is, of course, standard scientific practice. When derived from a well-articulated theory, a conditional submodel with z_t super exogenous seems to satisfy the requirement for Zellner causality of “predictability according to a law” (see Zellner [45]).

2.7. Comments

The motivation for introducing the concept of weak exogeneity is that it provides a sufficient⁹ condition for conducting inference conditionally on z_t without loss of relevant sample information. Our concept is a direct extension of Koopmans’ [21] discussion of exogeneity. He shows that an implicit static simultaneous equations system which has the properties: (a) the variables of the first block of equations do not enter the second block, (b) the disturbances between the two blocks are independent, and (c) the Jacobian of the transformation from the disturbances to the observables is nowhere zero, will have a likelihood function which factors into two components as in (18), a conditional and a marginal. The variables in the second block are labeled exogenous. Implicit in his analysis is the notion that the parameters of interest are all located in the first block and that this parameterization operates a cut. The failure to state precisely these components of the definition, leads to a lack of force in the definition as is illustrated in several of the examples in this paper. Koopmans then analyzes dynamic systems in the same framework leading to a notion of exogeneity which corresponds to our strong exogeneity and predeterminedness corresponding to that concept as defined above.

Koopmans presents sufficient conditions for the factorization of the likelihood but does not discuss the case where the factorization holds but his sufficient conditions do not. Our work therefore extends Koopmans’ by making precise the assumptions about the parameters of interest and by putting the definitions squarely on the appropriate factorization of the likelihood. More recent literature has in fact stepped back from Koopmans’ approach, employing definitions such as that of strict exogeneity in Section 2.2. As shown in Section 4, strict exogeneity, when applied to dynamic simultaneous equations models includes condition

⁹It is also necessary for most purposes. However, since in (14) ψ need not depend on all the elements in λ_1 it might happen that ψ and λ_2 are variation free even though λ_1 and λ_2 are not in which case neglecting the restrictions between λ_1 and λ_2 *might* entail no loss of efficiency for inference on ψ . More subtly, whether or not cuts are necessary to conduct inference based on partial models without loss of information obviously very much depends on how sample information is measured. See in particular the concepts of G - and M -ancillarity in Barndorff-Nielsen [1].

(iii) of Definition 2.6 together with predeterminedness; condition (ii) of Definition 2.5 is not required explicitly but, at least for just identified models, is often satisfied by construction; condition (i) of Definition 2.5 is certainly absent which, in our view, is a major lacuna¹⁰ since, unless it holds, strict exogeneity of z_t does not ensure that there is no loss of relevant sample information when conducting inference conditionally on z_t . On the other hand, if (i) and (ii) hold, then (iii) becomes irrelevant¹¹ since it no longer affects inference on the parameters of interest. This does not mean that condition (iii) has no merit on its own—a model user might express specific interest in detecting causal orderings and ψ should then be defined accordingly—but simply that it is misleading to emphasize Granger noncausality when discussing exogeneity. The two concepts serve different purposes: weak exogeneity validates conducting *inference* conditional on z_t while Granger noncausality validates *forecasting* z and then forecasting y conditional on the future z 's. As is well known, the condition that y does not Granger cause z is neither necessary nor sufficient for the weak exogeneity of z . Obviously, if estimation is required before conditional predictions are made, then strong exogeneity which covers both Granger noncausality *and* weak exogeneity becomes the relevant concept.

Note that if $[(y_t | z_t; \lambda_1), (z_t; \lambda_2)]$ operates a sequential cut, then the information matrix, if it exists, is block-diagonal between λ_1 and λ_2 . In fact for most of the examples discussed in this paper and in EHR the condition that the information matrix be block-diagonal appears to be equivalent to the condition that the parameterization should operate a sequential cut. However, at a more general level, the finding that the information matrix is block-diagonal between two sets of parameters, one of which contains all the parameters of interest, does not entail that the likelihood function factorize as in (18). Block-diagonality of the information matrix may reflect other features of the likelihood function. Therefore, it seems difficult to discuss exogeneity by means of information matrices without explicitly referring to reparameterizations in terms of conditional and marginal submodels. Further, information matrices are often difficult to obtain analytically especially in the presence of lagged endogenous variables.

Note also that some definitions seem designed to validate specific estimation methods such as ordinary least squares within a single equation framework. For example, Phillips [29, Section IV] presents conditions justifying least squares estimation in dynamic systems, which if fulfilled would allow regressors to be treated as “given,” despite the presence of Granger causal feedbacks. The concept of weak exogeneity is *not directly* related to validating specific estimation methods but concerns instead the conditions under which attention may be restricted to conditional submodels without loss of relevant sample information.

¹⁰This criticism is hardly specific to the concept of exogeneity. For example, unless there are parameters of interest, it is meaningless to require that an estimator should be consistent since it is always possible to redefine the “parameters” such that any chosen convergent estimation method yields consistent estimates thereof (see e.g., Hendry [17]).

¹¹Evidently if one wished to test the conditions under which (ii) held then overidentifying restrictions such as the ones typically implied by Granger noncausality would affect the properties of the test.

Later selection of an *inappropriate* estimator may produce inefficiency (and inconsistency) even when weak exogeneity conditions are fulfilled.

Many existing definitions of exogeneity have been formulated in terms of orthogonality conditions between observed variables and (unobservable) disturbances in linear relationships within processes which are usually required to be Gaussian. Definitions 2.5 and 2.6 apply equally well to any joint density function and therefore encompass nonlinear and non-Gaussian processes and truncated or otherwise limited dependent variables. As such nonclassical models come into more use it is particularly important to have definitions of exogeneity which can be directly applied. See for example Gouriéroux, et al. [15] or Maddala and Lee [25]. For a formulation tantamount to weak exogeneity in the context of conditional logit models, see McFadden [24, Section 5.1]. Exogeneity also has been discussed from the Bayesian point of view by Florens and Mouchart [10]. The issue then becomes whether or not the posterior density of the parameters of interest as derived from a conditional submodel coincides with that derived from the complete model. Such is the case if z_t is weakly exogenous and in addition λ_1 and λ_2 in Definition 2.5 are a priori independent. However the conditions are not necessary and it may be the case that, in the absence of a sequential cut, the prior density is such that the desired result is still achieved.

3. EXAMPLES

Many of the points made in the previous section can be illustrated with the simplest of all multivariate models, the bivariate normal. Because this is a static model, the concepts of weak and strong exogeneity coincide as do the concepts of predeterminedness and strict exogeneity. The central role of the choice of parameters of interest is seen directly.

EXAMPLE 3.1: Let the data on y_t and z_t be generated by:

$$(24) \quad \begin{bmatrix} y_t \\ z_t \end{bmatrix} \sim IN(\mu, \Omega), \quad \mu = (\mu_i), \quad \Omega = (\omega_{ij}), \quad i, j = 1, 2,$$

with the conditional distribution of y_t given z_t :

$$(25) \quad y_t | z_t \sim IN(\alpha + \beta z_t, \sigma^2)$$

where $\beta = \omega_{12}/\omega_{22}$, $\alpha = \mu_1 - \beta\mu_2$, and $\sigma^2 = \omega_{11} - \omega_{12}^2/\omega_{22}$. Letting

$$(26) \quad u_{1t} = y_t - E(y_t | z_t), \quad v_{2t} = z_t - E(z_t),$$

the model is correspondingly reformulated as

$$(27) \quad y_t = \alpha + \beta z_t + u_{1t}, \quad u_{1t} \sim IN(0, \sigma^2),$$

$$(28) \quad z_t = \mu_2 + v_{2t}, \quad v_{2t} \sim IN(0, \omega_{22}),$$

where $\text{cov}(z_t, u_{1t}) = \text{cov}(v_{2t}, u_{1t}) = 0$ by construction. The parameters of the conditional model (27) are $(\alpha, \beta, \sigma^2)$ and those of the marginal model (28) are

(μ_2, ω_{22}) . They are in one-to-one correspondence with (μ, Ω) and are variation-free since, for arbitrary choices of $(\alpha, \beta, \sigma^2)$ and (μ_2, ω_{22}) in their sets of admissible values which are respectively $R^2 \times R_+$ and $R \times R_+$, μ and Ω are given by

$$(29) \quad \mu = \begin{bmatrix} \alpha + \beta\mu_2 \\ \mu_2 \end{bmatrix}, \quad \Omega = \begin{bmatrix} \sigma^2 + \beta^2\omega_{22} & \beta\omega_{22} \\ \beta\omega_{22} & \omega_{22} \end{bmatrix},$$

and the constraint that Ω be positive definite is automatically satisfied (see Lemma 5.1 in Drèze and Richard [5] for a generalization of this result to multivariate regression models). It follows that z_t is weakly exogenous for $(\alpha, \beta, \sigma^2)$ or for any well-defined function thereof.

However, similar reasoning applies by symmetry to the factorization

$$(30) \quad z_t = \gamma + \delta y_t + u_{2t}, \quad u_{2t} \sim IN(0, \tau^2),$$

$$(31) \quad y_t = \mu_1 + v_{1t}, \quad v_{1t} \sim IN(0, \omega_{11}),$$

where $\delta = \omega_{12}/\omega_{11}$, $\gamma = \mu_2 - \delta\mu_1$, $\tau^2 = \omega_{22} - \omega_{12}^2/\omega_{11}$, and $\text{cov}(y_t, u_{2t}) = \text{cov}(v_{1t}, u_{2t}) = 0$ by construction. Therefore, y_t is weakly exogenous for (γ, δ, τ^2) or for any well-defined function thereof. In this example the choice of parameters of interest is the sole determinant of weak exogeneity which is, therefore, not directly testable.

Next, consider the concept of predeterminedness which is here equivalent to that of strict exogeneity. Regardless of the parameters of interest, z_t is predetermined in (27) by construction and so is y_t in (30). Which variable is predetermined depends upon the form of the equation, not upon the properties of the joint density function. Until some of the parameters are assumed to be more fundamental or structural (i.e., parameters of interest), the notion of predeterminedness has no force. When δ is the parameter of interest, z_t is predetermined in equation (27) but not weakly exogenous while y_t is weakly exogenous but not predetermined. Similar results hold in more complex models where the assumptions of exogeneity can be tested.

This example also illustrates the ambiguity in Koopmans' sufficient conditions as discussed in Section 2.7 since their application leads to the conclusion that z_t is exogenous in (27) and (28) while y_t is exogenous in (30) and (31), a conclusion which seems to misrepresent Koopmans' views about exogeneity.

Now consider the concepts of structural invariance and super exogeneity. Will the parameter β in (27) be invariant to an intervention which changes the variance of z ? The answer depends upon the structure of the process. If β is truly a constant parameter (because, e.g., (27) is an autonomous behavioral equation) then σ_{12} will vary with σ_{22} since, given (26), $\sigma_{12} = \beta\sigma_{22}$. Alternatively it might be σ_{12} which is the fixed constant of nature in (24) and in this case β will not be invariant to changes in σ_{22} ; z_t can be weakly exogenous for β within one regime with β a derived parameter which changes between regimes. By making β the parameter of interest, most investigators are implicitly assuming that it will

remain constant when the distribution of the exogenous variables changes; however, this is an assumption which may not be acceptable in the light of the Lucas [23] critique. Similar arguments apply to α or σ^2 . Therefore, if $(\alpha, \beta, \sigma^2)$ are invariant to any changes in the distribution of z_t or, more specifically in this restricted framework, to changes in μ_2 and ω_{22} , then z_t is super exogenous for $(\alpha, \beta, \sigma^2)$. If, on the other hand, β is invariant to such changes while α and σ^2 are not, e.g., because μ_1 and ω_{11} are invariant, then z_t might be weakly exogenous for β within each regime but is not super exogenous for β since the marginal process (28) now contains valuable information on the shifts of α and σ^2 between regimes.¹² It is clear from the above argument that weak exogeneity does not imply structural invariance. It is also clear that even if β is invariant to changes in the distribution of z or in fact the conditional model (27) is structurally invariant, the parameter of interest could be γ and therefore z_t would not be weakly exogenous, and thus not super exogenous either.

Finally, since weak exogeneity explicitly requires that all relevant sample information be processed, overidentifying restrictions are bound to play an essential role in a discussion of weak exogeneity assumptions.¹³ This will be discussed further in Section 4 within the framework of dynamic simultaneous equations models. Example 3.2 illustrates the role of overidentifying restrictions in a simple structure.

EXAMPLE 3.2: Consider the following two-equation overidentified model:

$$(32) \quad y_t = z_t \beta + \epsilon_{1t},$$

$$(33) \quad z_t = z_{t-1} \delta_1 + y_{t-1} \delta_2 + \epsilon_{2t},$$

$$(34) \quad \begin{bmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{bmatrix} \sim IN(0, \Sigma), \quad \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix}.$$

Equation (33) is a typical control rule for an agent attempting to control y . For example, this could be a governmental policy reaction function or a farmer's supply decision or a worker's rule for deciding whether to undertake training. These cobweb models have a long history in econometrics. The parameter of interest is assumed to be β .

¹²This illustrates the importance of incorporating in Definition 2.9 the requirement that the conditional model $D(y_t | z_t, X_{t-1}, \lambda_1)$ be structurally invariant even though ψ may depend only on a subvector of λ_1 .

¹³An interesting example of the complexities arising from overidentification occurs if $\omega_{11} = 1$ in (24) a priori. Then the factorization (27) and (28) no longer operates a cut as a result of the overidentifying constraint $\sigma^2 + \beta^2 \omega_{22} = 1$, while the factorization (30) and (31) still does. Further, β and σ^2 are well-defined functions of (γ, δ, τ^2) since $\beta = \delta / (\delta^2 + \tau^2)$ and $\sigma^2 = \tau^2 / (\delta^2 + \tau^2)$ while α is not. Therefore, z_t is no longer weakly exogenous for (β, σ^2) while y_t now is! Neither of these two variables is weakly exogenous for α .

The reduced form consists of (33) and

$$(35) \quad y_t = \beta \delta_1 z_{t-1} + \beta \delta_2 y_{t-1} + v_t$$

$$(36) \quad \begin{bmatrix} v_t \\ \epsilon_{2t} \end{bmatrix} \sim IN(0, \Omega), \quad \Omega = \begin{bmatrix} \sigma_{11} + 2\beta\sigma_{12} + \beta^2\sigma_{22} & \sigma_{12} + \beta\sigma_{22} \\ \sigma_{12} + \beta\sigma_{22} & \sigma_{22} \end{bmatrix},$$

and the conditional density of y_t given z_t is

$$(37) \quad D(y_t | z_t, X_{t-1}, \theta) = N(bz_t + c_1 z_{t-1} + c_2 y_{t-1}, \sigma^2) \quad \text{where}$$

$$(38) \quad b = \beta + \frac{\sigma_{12}}{\sigma_{22}}, \quad c_1 = -\delta_1 \frac{\sigma_{12}}{\sigma_{22}}, \quad \sigma^2 = \sigma_{11} - \sigma_{12}^2 / \sigma_{22},$$

which can be written as the regression

$$(39) \quad y_t = bz_t + c_1 z_{t-1} + c_2 y_{t-1} + u_t, \quad u_t \sim IN(0, \sigma^2).$$

The condition which is of first concern is the value of the parameter σ_{12} . If $\sigma_{12} = 0$, then z_t is predetermined in (32) and is weakly exogenous for β since (β, σ_{11}) and $(\delta_1, \delta_2, \sigma_{22})$ operates a cut. Even so, for $\delta_2 \neq 0$, y Granger causes z and therefore z is not strongly exogenous, nor is it strictly exogenous. However, the important criterion for efficient estimation is weak exogeneity, not strong exogeneity, and tests for Granger causality have no bearing on either the estimability of (32), or the choice of estimator.

If σ_{12} is not zero, then z_t is not weakly exogenous for β because this parameter cannot be recovered from only the parameters b, c_1, c_2, σ^2 of the conditional distribution (37). In (32) z_t is also not predetermined; however, in (39) it is, again showing the ambiguities in this concept. Whether or not a variable is predetermined depends on which equation is checked, and is not an intrinsic property of a variable.

The preceding results remain unchanged if $\delta_2 = 0$, in which case y does not Granger cause z , yet z_t is still not weakly exogenous for β when $\sigma_{12} \neq 0$. Granger noncausality is neither necessary nor sufficient for weak exogeneity, or for that matter, for predeterminedness.

Suppose instead that b is the parameter of interest and $\delta_2 \neq 0$. Then OLS on (39) will give a consistent estimate. This however will not be an efficient estimate since the parameters should satisfy the restriction

$$(40) \quad \delta_1 c_2 = \delta_2 c_1$$

and consequently joint estimation of (39) and (33) would be more efficient. The parameterization (b, c_1, c_2, σ^2) , $(\delta_1, \delta_2, \sigma_{22})$ does not operate a cut because the parameter sets are not variation free so z_t is not weakly exogenous for b . If, however, $\delta_2 = 0$ so that the system becomes just identified then z_t will be weakly exogenous for b as (b, c_1, σ^2) , (δ_1, σ_{22}) operates a cut. In both cases, z_t is still predetermined in (39).

Which parameter “ought” to be the parameter of interest requires further information about the behavior of the system and its possible invariants. Usually, it seems desirable to choose as parameters of interest, those parameters which are invariant to changes in the distribution of the weakly exogenous variables. Returning to the first case where β is the parameter of interest and $\sigma_{12} = 0$, the investigator might assume that (β, σ_{11}) would be invariant to changes in the distribution of z . If this were valid, z_t would be super exogenous, even though it is still Granger caused by y so it is not strongly exogenous nor strictly exogenous. Changes in the parameters of (33) or even of the distribution of z_t will not affect estimation of β nor will control of z affect the conditional relation between y_t and z_t given in (32). Conversely, if $\delta_2 = 0$, but $\sigma_{12} \neq 0$, then (b, c_1, σ^2) and (δ_1, σ_{22}) operates a cut, and z is strictly exogenous in (39) and strongly exogenous for b , yet that regression is by hypothesis not invariant to changes in either δ_1 or σ_{22} , cautioning against constructing cuts which do not isolate invariants.

The assumption of super exogeneity is testable if it is known that the parameters of the marginal distribution have changed over the sample period. A test for changes in β could be interpreted as a test for super exogeneity with respect to the particular interventions observed.

To clarify the question of structural invariance in this example, consider a derivation of the behavioral equation (32) based on the assumption that the agent chooses y to maximize his expected utility conditional on the information available to him. Let the utility function be

$$(41) \quad U(y, z; \beta) = -(y - z\beta)^2$$

where β is a parameter which is by hypothesis completely unrelated to the distribution of z and hence is invariant to any changes in the δ 's in equation (33). Allowing for a possible random error v_t , arising from optimization, the decision rule is

$$(42) \quad y_t = \beta z_t^e + v_t$$

where z_t^e represents the agent's expectation of z_t conditionally on his information set I_t . In the perfect information case where z_t is contained in I_t , $z_t^e = z_t$ and (32) follows directly from (42). Hence β is structurally invariant and the assumption that $\sigma_{12} = \text{cov}(v_t, \epsilon_{2t}) = 0$ is sufficient for the weak exogeneity of β and, consequently, for its super exogeneity. The imperfect information case raises more subtle issues since, as argued e.g., in Hendry and Richard [19], z_t^e may not coincide with the expectation of z_t as derived from (33). In this example, however, we discuss only the rational expectations formulation originally proposed by Muth [27] whereby it is assumed that z_t^e and $E(z_t | \cdot)$ in (33) coincide. Hence (32) follows from (42) and

$$(43) \quad \epsilon_{1t} = v_t - \beta \epsilon_{2t}$$

so that $\sigma_{12} = \text{cov}(v_t, \epsilon_{2t}) - \beta \sigma_{22}$. Therefore, the conventional assumption that $\text{cov}(v_t, \epsilon_{2t}) = 0$ entails that $\sigma_{12} = -\beta \sigma_{22} \neq 0$ in which case z_t is neither weakly

exogenous nor super exogenous for β even though β is invariant. On the other hand, rational expectations per se does not exclude the possibility that $\sigma_{12} = 0$ (so that z_t remains weakly exogenous for β) since, e.g.,

$$(44) \quad \text{cov}(v_t, \epsilon_{2t}) = \sigma_{22} \beta$$

suffices.

Under the familiar assumptions, $\text{cov}(v_t, \epsilon_{2t}) = 0$, the conditional expectation (37), and the reduced form, (35), coincide. No current value of z belongs in the conditional expectation of y_t given (z_t, I_t) . Nevertheless, z_t is not weakly exogenous for β because the parameter β cannot be recovered from the reduced form coefficients c_1 and c_2 alone. This illustrates that even when the current value fails to enter the conditional expectation, weak exogeneity need not hold.

If the c_i were the parameters of interest, then z_t would be weakly exogenous, but these reduced form parameters are not structurally invariant to changes in the δ 's. The Lucas [23] criticism applies directly to this equation regardless of whether y Granger causes z . The derivation and the noninvariance of these parameters suggests why they should not be the parameters of interest. Once again, testing for Granger causality has little to do with the Lucas criticism or the estimability or formulation of the parameters of interest. It is still possible to estimate β efficiently, for example by estimating (32) and (33) jointly as suggested by Wallis [41], but this requires specifying and estimating both equations.¹⁴ If there is a structural shift in the parameters of the second equation, this must also be allowed for in the joint estimation. This example shows the close relationship between weak exogeneity and structural invariance and points out how models derived from rational expectations behavior may or may not have weak exogeneity and structural invariance.

EXAMPLE 3.3: This final example shows that with a slight extension of the linear Gaussian structure to include serial correlation, the concept of predeterminedness becomes even less useful.

Consider the model

$$(45) \quad y_t = \beta z_t + u_t,$$

$$(46) \quad u_t = \rho u_{t-1} + \epsilon_{1t},$$

$$(47) \quad z_t = \gamma y_{t-1} + \epsilon_{2t},$$

$$(48) \quad \begin{bmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{bmatrix} \sim IN(0, \Sigma), \quad \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix}.$$

Although this model is unidentified in a rather subtle sense, this need not concern us here as all the special cases to be discussed will be identified. The issue is dealt with more fully in EHR.

¹⁴Depending on the model formulation, instrumental variables estimation of (say) (32) alone is sometimes fully efficient.

The conditional expectation of y_t given z_t and X_{t-1} implies the regression

$$(49) \quad y_t = bz_t + cy_{t-1} + dz_{t-1} + \eta_t, \quad \eta_t \sim IN(0, \sigma^2),$$

where

$$(50) \quad b = \beta + \sigma_{12}/\sigma_{22}, \quad c = \rho - \gamma\sigma_{12}/\sigma_{22}, \quad d = -\beta\rho, \quad \sigma^2 = \sigma_{11} - \sigma_{12}^2/\sigma_{22}.$$

The covariance between z_t and u_t is given by

$$(51) \quad \text{cov}(z_t, u_t) = \left(\sigma_{12} + \sigma_{11} \frac{\gamma\rho}{1 - \rho^2} \right) / (1 - \gamma\rho\beta).$$

Note first that, as indicated by (51), the condition $\sigma_{12} = 0$ is *not* sufficient for the predeterminedness of z_t in (45). However, $\sigma_{12} = 0$ is sufficient for the weak exogeneity of z_t for the parameters β and ρ , as can be seen directly from (50) where the parameters of the conditional model (49) are subject to a common factor restriction but are variation free with those of the marginal model (47). Thus, the parameters of (49) could be estimated by imposing the restrictions through some form of autoregressive maximum likelihood method. Ordinary least squares estimation of β in (45) will be inconsistent whereas autoregressive least squares will be both consistent and asymptotically efficient. This example shows the advantages of formulating definitions in terms of expectations conditional on the past.

A second interesting property of this model occurs when $\sigma_{12} \neq 0$ but $\gamma = 0$. Again (51) shows that z_t is not predetermined in (45), but surprisingly, it is weakly exogenous for β and ρ . The three regression coefficients in (49) are now a nonsingular transformation of the three unknown parameters ($\beta, \rho, \sigma_{12}/\sigma_{22}$) and these operate a cut with respect to the remaining nuisance parameter σ_{22} . Ordinary least squares estimation of (49) provides efficient estimates of its parameters and the maximum likelihood estimate of β is $-d/c$. Both ordinary least squares and autoregressive least squares estimation of (45) would yield inconsistent estimates of β .

The case where

$$(52) \quad (1 - \rho^2)\sigma_{12} + \gamma\rho\sigma_{11} = 0$$

raises several important issues which are discussed in detail in EHR. In short, the condition (52) identifies the model but violates both conditions (i) and (ii) in Definition 2.5 so that z_t is not weakly exogenous for (β, ρ) , neither is it for (b, c, d) in (49). In particular, the autoregressive least squares estimator of β in (45) and (46) is inconsistent even though, as a consequence of the predeterminedness of z_t in (45), the first step ordinary least squares estimators of β in (45) and ρ in (46) are consistent (but not efficient).

This concludes the discussion of the examples. It is hoped that these have shown the usefulness of the concepts of weak and strong exogeneity, structural invariance, and super exogeneity in analyzing familiar and possibly some unfa-

miliar situations. Further examples can be found in EHR including a truncated latent variable model based upon Maddala and Lee [25].

4. APPLICATION TO DYNAMIC SIMULTANEOUS EQUATIONS MODELS

In this section we shall apply our analysis to dynamic simultaneous equations models (DSEM). As this is the arena in which notions of exogeneity are most heavily used and tested, it is important to relate our concepts to conventional wisdom. It will be shown that the conventional definitions must be supplemented with several conditions for the concepts to have force. However, when these conditions are added, then in standard textbook models, predeterminedness becomes equivalent to weak exogeneity and strict exogeneity becomes equivalent to strong exogeneity. Finally, our framework helps clarify the connections between such (modified) concepts and the notions of Wold causal orderings (see Strotz and Wold [40]), “block recursive structures” (see Fisher [9]), and “exogeneity tests” as in Wu [44].

Following Richard [31, 32] the system of equations need not be complete and thus the analysis is directly a generalization of the conventional DSEM. Assuming normality and linearity of the conditional expectations $\phi_t = E(x_t | X_{t-1}, \theta)$, let¹⁵

$$(53) \quad D(x_t | X_{t-1}, \theta) = f_N^n(x_t | \sum \Pi(i)x_{t-i}, \Omega)$$

where $\{\Pi(i)\}$ and Ω are functions of a vector of unknown parameters $\theta \in \Theta$.

Define the “innovations” or “reduced form disturbances” v_t by

$$(54) \quad v_t = x_t - \phi_t = x_t - \sum \Pi(i)x_{t-i}.$$

Then, ϕ_t being conditional on X_{t-1} ,

$$(55) \quad \text{cov}(v_t, x_{t-i}) = 0 \quad \text{for all } i > 0, \quad \text{and hence}$$

$$(56) \quad \text{cov}(v_t, v_{t-i}) = 0 \quad \text{for all } i > 0.$$

We define the dynamic multipliers $Q(i)$ by the recursion

$$(57) \quad \begin{aligned} Q(0) &= I_n \quad \text{and} \\ Q(i) &= \sum_{j=1}^i \Pi(j)Q(i-j), \quad \text{for all } i \geq 1, \end{aligned}$$

¹⁵Our framework explicitly requires that the distribution of the endogenous variables be completely specified. Normality (and linearity) assumptions are introduced here because they prove algebraically convenient. Other distributional assumptions could be considered at the cost of complicating the algebra. Furthermore, there exist distributions, such as the multivariate student distribution, for which there exist no cuts. Evidently weak exogeneity can always be achieved by construction, simply by specifying independently of each other a conditional and a marginal model, but is then no longer testable. More interestingly, conditions such as the ones which are derived below could be viewed as “approximate” or “local” exogeneity conditions under more general specifications. Given the recent upsurge of nonlinear non-Gaussian models in econometrics this is clearly an area which deserves further investigation.

and note that

$$(58) \quad \text{cov}(v_t, x_{t+i}) = Q(i)\Omega \quad \text{for all } i \geq 0.$$

Often the specification of θ is (partially) achieved by considering sets of behavioral relationships. Such relationships can correspond to optimizing behavior given expectations about future events, allow for adaptive responses and include mechanisms for correcting previous mistakes. In our framework, where attention is focussed on the conditional densities $D(x_t | X_{t-1}, \theta)$ it is natural to specify these relationships in terms of the conditional expectations ϕ_t . Consider, therefore, a set of $g \leq n$ linear behavioral relationships of the form

$$(59) \quad B\phi_t + \sum C(i)x_{t-i} = 0$$

where B and $\{C(i)\}$ are $g \times n$ matrix functions of a vector of "structural" coefficients $\delta \in \Delta$, with rank $B = g$ almost everywhere in Δ . The δ 's are typically parameters of interest. We can also define a g -dimensional vector of unobservable "structural disturbances:"

$$(60) \quad \epsilon_t = Bx_t + \sum C(i)x_{t-i}$$

which also satisfy, by construction, the properties (55), (56).

Let Σ denote the covariance matrix of ϵ_t . In all generality Σ is also treated as a function of δ . From (53), (54), (59), and (60) we must have $\epsilon_t = Bv_t$ and

$$(61) \quad B\Pi(i) + C(i) \equiv 0, \quad \text{for all } i \geq 0; \quad \Sigma = B\Omega B'.$$

The identities (61) define a correspondence between Δ and Θ or, equivalently, a function h from Δ to $P(\Theta)$, the set of all subsets of Θ . To any given $\delta \in \Delta$, h associates a subset of Θ which we denote by $h(\delta)$. In the rest of the paper it is assumed that: (i) δ is *identified* in the sense that

$$(62) \quad \text{for all } \delta, \delta^* \in \Delta, \quad \delta \neq \delta^* \rightarrow h(\delta) \cap h(\delta^*) = \emptyset;$$

(ii) all values in Θ are compatible with (61),

$$(63) \quad \Theta = \bigcup_{\delta \in \Delta} h(\delta),$$

so that $\{h(\delta)\}$ is a partition of Θ .

Let s denote the number of nonzero columns in $\{\Pi(i)\}$ and C_n the set of $n \times n$ symmetric positive definite matrices. If $\Theta = R^s \times C_n$, except for the set of zero Lebesgue measure, then the model (53) is *just identified*. It is *overidentified* if Θ is a strict subset of $R^s \times C_n$.

When $g < n$, it often proves convenient to define an auxiliary parameter vector, say $\bar{\theta} \in \bar{\Theta}$ of the form $\bar{\theta} = (\delta, \theta_2)$ where θ_2 is a subvector of θ defined in such a way that $\bar{\Theta}$ and Θ are in one-to-one correspondence. If, in particular, $\{\Pi(i)\}$ and Ω are subject to no other constraints than those derived from the identities (61) as implicitly assumed in this section, then we can select for θ_2 the

coefficients of $n - g$ unconstrained “reduced form” equations,¹⁶ whereby

$$(64) \quad \bar{\Theta} = \Delta \times \Theta_2, \quad \text{with} \quad \Theta_2 = R^{s(n-g)} \times C_{n-g}.$$

The specification of many econometric models “allows for” serial correlation of the residuals, i.e., incorporates linear relationships of the form

$$(65) \quad B^*x_t + \sum C^*(i)x_{t-i} = u_t \sim N(0, \Phi)$$

where B^* and $\{C^*(i)\}$ are claimed to be parameters of interest (or well defined functions thereof) and u_t is seen as a g -dimensional “autonomous” process, subject to serial correlation. Note that if (65) is to be used to *derive* the distribution of x_t from that of u_t , then the system must be “complete”, i.e., $g = n$.

Provided u_t has an autoregressive representation,

$$(66) \quad u_t = \sum R_i u_{t-i} + e_t, \quad \text{where} \quad e_t \sim IN(0, \Sigma),$$

then (65) can be transformed to have serially uncorrelated “errors” (the new parameterization being subject to common factor restrictions as in Sargan [37]) in which case the transformed model can be reinterpreted in terms of conditional expectations as in (53). More general specifications of u_t are not ruled out in principle, but might seriously complicate the analysis.

We can now unambiguously characterize and inter-relate the concepts of Granger noncausality, predeterminedness and strict exogeneity, as given in Definitions 2.1–2.3, for potentially overidentified and incomplete DSEM which have been transformed to have serially uncorrelated residuals. Since these concepts may apply only to a subset of the equation system (59), this is accordingly partitioned into the first $g_1 \leq p$ equations and the remaining $g_2 = g - g_1 \leq q$ equations—see e.g. Fisher [9] on the notion of block recursive structures. We partition the Π ’s, Q ’s, and Ω conformably with the variables $x'_t = (y'_t z'_t)$, B conformably with the variables and the equations and the C ’s and Σ conformably with the equations as:

$$(67) \quad \begin{aligned} \Pi(i) &= \begin{bmatrix} \Pi_1(i) \\ \Pi_2(i) \end{bmatrix} = \begin{bmatrix} \Pi_{11}(i) & \Pi_{12}(i) \\ \Pi_{21}(i) & \Pi_{22}(i) \end{bmatrix}, \\ Q(i) &= \begin{bmatrix} Q_1(i) \\ Q_2(i) \end{bmatrix} = \begin{bmatrix} Q_{11}(i) & Q_{12}(i) \\ Q_{21}(i) & Q_{22}(i) \end{bmatrix}, \\ \Omega &= (\Omega_1 \Omega_2) = \begin{bmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{21} & \Omega_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}, \\ C(i) &= \begin{bmatrix} C_1(i) \\ C_2(i) \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}. \end{aligned}$$

¹⁶This is current practice in the literature on so-called limited information procedures. Non-Bayesian inference procedures based on likelihood principles are invariant with respect to the choice

THEOREM 4.1: *For the class of models defined by (53) plus (59): (i) y does not Granger cause z if and only if*

$$Q_{21}(i) = 0, \quad \text{for all } i \geq 1;$$

(ii) z_t is predetermined in the first g_1 equations of (59) if and only if

$$B_1 \Omega_2 = 0;$$

(iii) z_t is strictly exogenous in the first g_1 equations of (59) if and only if

$$B_1 \Omega Q'_2(i) = 0, \quad \text{for all } i \geq 0.$$

(iv) Conditions (i) and (ii) are sufficient for (iii). If $g_1 = p$, they are also necessary for (iii).

(v) If $B_{21} = 0$, $\Sigma_{12} = 0$, and rank $B_{22} = q (= g_2)$, then z_t is predetermined in the first g_1 equations of (59).

PROOF: The proof follows from the Definitions 2.1–2.3 together with (57), wherefrom it can be shown by recurrence that $(\Pi_{21}(i) = 0; i \geq 1)$ is equivalent to $(Q_{21}(i) = 0; i \geq 1)$. See EHR for more details.

In order to discuss weak exogeneity the parameters of interest must be defined. In the theorems below it will be assumed that the parameters of interest are all grouped together in the first g_1 equations. Thus it is not a cavalier matter which equations are put in the first group. For example, in a control problem, the first g_1 equations might describe the behavior of the economic agents given the controlled values of z_t , while the remaining g_2 equations describe the control rules which have been operative.

Factorizing the joint density (53) also requires the introduction of an appropriate reparameterization. This is the object of Lemma 4.2 which translates into our notation results which are otherwise well-known.

LEMMA 4.2: *The joint density (53) factorizes into the product of the conditional density*

$$(68) \quad D(y_t | z_t, X_{t-1}, \lambda_1) = f_N^q(y_t | \Delta_{12} z_t + \sum \Pi_{1,2}(i) x_{t-i}, \Omega_{11,2})$$

and the marginal density

$$(69) \quad D(z_t | X_{t-1}, \lambda_2) = f_N^q(z_t | \sum \Pi_2(i) x_{t-i}, \Omega_{22})$$

with $\lambda_1 = (\Delta_{12}, \{\Pi_{1,2}(i)\}, \Omega_{11,2})$, $\lambda_2 = (\{\Pi_2(i)\}, \Omega_{22})$,

$$(70) \quad \Delta_{12} = \Omega_{12} \Omega_{22}^{-1}, \quad \Omega_{11,2} = \Omega_{11} - \Omega_{12} \Omega_{22}^{-1} \Omega_{21}, \text{ and } \Pi_{1,2}(i) = \Pi_1(i) - \Delta_{12} \Pi_2(i).$$

PROOF: See e.g. Press [30, Sections 3.4 and 3.5].

of these $n - g$ reduced form equations, provided they form a nonsingular set of equations together with the g structural relationships (59). Also, in a Bayesian framework there exist prior densities on θ such that the corresponding posterior densities on δ have similar invariance properties. For details, see e.g. Dreze and Richard [5] for $g = 1$, or Richard [31] for $g > 1$.

If the model (53) is *just identified*, then λ_1 and λ_2 are variation free with respective domains of variation $\Lambda_1 = R^{p \times q} \times \{R^{p \times n}\} \times C_p$ and $\Lambda_2 = \{R^{q \times n}\} \times C_q$ and z_t is weakly exogenous for ψ if and only if ψ is a function of λ_1 only. However, in order to be operational within the framework of DSEM's, such a condition should be expressed in terms of the structural coefficients δ since these are themselves typically parameters of interest. Also, most applications involve overidentified models for which λ_1 and λ_2 are no longer variation free unless some additional conditions are satisfied. Thus, the object of Theorem 4.3 is to derive *general* conditions on δ for the weak exogeneity of z_t for ψ . By their nature, these conditions are *sufficient* and, as in Section 3, it is easy to construct examples in which they are *not necessary*. Consequently, insofar as so-called "exogeneity tests" are typically tests for such conditions, rejection on such a test does not necessarily entail that the weak exogeneity assumption is invalid (see e.g. Example 3.3 when $\sigma_{12} \neq 0$ and $\gamma = 0$).

THEOREM 4.3: *For the DSEM in (53) plus (59) consider the following conditions:*

- (i) $B_1 \Omega_2 = 0$,
- (ii) $B_{21} = 0$,
- (iii) $(B_1, \{C_1(i)\}, \Sigma_{11})$ and $(B_2, \{C_2(i)\}, \Sigma_{22})$ are variation free,
- (iv) ψ is a function of $(B_1, \{C_1(i)\}, \Sigma_{11})$,
- (v) $\Sigma_{12} = 0$,
- (vi) $\text{rank } B_{22} = q$,
- (vii) $(B_2, \{C_2(i)\}, \Sigma_{22})$ are just identified parameters.

The following sets of conditions are sufficient for the weak exogeneity of z_t for ψ :

- (a) (i)(ii)(iii)(iv),
- (b) (ii)(iii)(iv)(v)(vi),
- (c) (i)(iii)(iv)(vii).

PROOF: The basic result (a) generalizes Theorem 3.1 in Richard [32] in that it also covers cases where restrictions are imposed on Σ . The proof in Richard extends to the more general case since, under (i) and (ii), the identity $\Sigma = B \Omega B'$ separates into the two identities $\Sigma_{11} = B_{11} \Omega_{11.2} B'_{11}$ and $\Sigma_{22} = B_{22} \Omega_{22} B'_{22}$. Result (b) follows from (a) together with condition (ii) and (v) in Theorem 4.1. Result (c) follows by applying (a) to a system consisting of the first g_1 behavioral relationships and g_2 *unrestricted* reduced form equations whose parameters are in one-to-one correspondence with $(B_2, \{C_2(i)\}, \Sigma_{22})$ and variation free with $(B_1, \{C_1(i)\}, \Sigma_{11})$ following conditions (vii) and (iii).

The major differences in the sufficient conditions for weak exogeneity and for

predeterminedness are conditions (iii) and (iv) of Theorem 4.3, which assure the model builder that there are no cross equation restrictions to the second block of equations and that there are no interesting parameters in that block.

To show the importance of these conditions in any definition, consider a set of $g \leq p < n$ just identified behavioral relationships, as given by (59) such that $B\Omega_2 \neq 0$. As is well known (see, for example, Strotz and Wold [40]) the system (59) can be replaced by an observationally equivalent one in which z_t is predetermined, and hence is strictly exogenous if y does not Granger cause z . For example let

$$(71) \quad \bar{B} = \Phi(I_g : -\Omega_{12}\Omega_{22}^{-1}),$$

$$(72) \quad \bar{C}(i) = -\bar{B}\Pi(i), \quad i \geq 1,$$

where Φ is an arbitrary but known $g \times g$ nonsingular matrix so that $(\bar{B}, \{\bar{C}(i)\})$ are just-identified by construction. Such transformations, with $\Phi = I_2$, have been implicitly used in the Examples 3.1–3.2. Replacing (59) by

$$(73) \quad \bar{B}\phi_t + \sum \bar{C}(i)x_{t-i} = 0$$

leaves (53) unaffected, but now $\bar{B}\Omega_2 = 0$. Consequently, $(\bar{B}, \{\bar{C}(i)\})$ can be estimated consistently from the conditional model $D(y_t | z_t, X_{t-1}, \cdot)$ together with (73). These estimates would be efficient provided (59) were just-identified. However, it is essential to realize that, since $g \leq p < n$, the parameters $(B, \{C(i)\})$ are typically not functions of $(\bar{B}, \{\bar{C}(i)\})$ alone and if the former are of interest, transforming (59) to (73) does not allow valid inference conditionally on z_t .

Thus, although at first sight, in normal DSEM weak exogeneity appears to be close to the notion of a Wold causal ordering, without the concept of parameters of interest the latter lacks force since there may be no cut which separates the parameters of interest and the nuisance parameters. Nevertheless, it must be stressed that Wold and Jureen [43, p. 14] *explicitly* include the condition that “each equation in the system expresses a unilateral causal dependence” which, in the spirit of our use of sequential cuts, seems designed to exclude arbitrary transformations of the system (59); see also the distinction in Bentzel and Hansen [2] between basic and derived models.

In Wu's [44] analysis, where $g_1 = 1$, it is implicit that conditions (iii) and (vii) of Theorem 4.3 are satisfied in which case the condition for predeterminedness ($B_1\Omega_2 = 0$) is indeed sufficient for the weak exogeneity of z_t for the parameters of the first behavioral equation (but not necessarily for other parameters of interest). It must be stressed, however, that if the remaining behavioral equations in the model under consideration are over-identified, then predeterminedness might no longer be sufficient on its own for the weak exogeneity of z_t . Therefore, even if the conditions (iii) and (iv) of Theorem 4.3 are incorporated in the definition of predeterminedness as is sometimes implicitly done, there would remain many situations where weak exogeneity and predeterminedness would still differ. Cases (a) and (b) in Theorem 4.3 provide *sufficient* conditions which

are applicable to more general cases than the one considered in Wu. Note, however, that condition (ii) in particular is not necessary and that case (c) could be made more general at the cost of some tedious notation as hinted by the following example.¹⁷

EXAMPLE 4.4: Consider a (complete) DSEM with $n = 3$, $p = g_1 = 1$, $q = g_2 = 2$, and

$$B = \begin{bmatrix} 1 & b_1 & 0 \\ b_2 & 1 & 0 \\ 0 & b_3 & 1 \end{bmatrix}, \quad C(1) = \begin{bmatrix} c_1 & 0 & 0 \\ 0 & c_2 & c_3 \\ 0 & c_4 & 0 \end{bmatrix}, \quad C(i) = 0, \quad i > 1.$$

The b 's and c 's are assumed to be variation free. The condition $B_1\Omega_2 = 0$, which is equivalent to $\sigma_{12} = b_2\sigma_{11}$ and $\sigma_{13} = 0$, is sufficient for the weak exogeneity of (y_{2t}, y_{3t}) for (b_1, c_1, σ_{11}) even though $B'_{21} = (b_2 0) \neq 0$ and the third behavioral relationship is overidentified (but does not contain y_{1t} !). Note that the predeterminedness of y_{2t} in the first behavioral relationship ($\sigma_{12} = b_2\sigma_{11}$) is sufficient for the *consistency* of OLS estimation of (b_1, c_1, σ_{11}) in that relationship but not for the weak exogeneity of (y_{2t}, y_{3t}) —or y_{2t} alone—for (b_1, c_1, σ_{11}) . In the absence of additional restrictions such as $\sigma_{13} = 0$ a more *efficient* estimator of (b_1, c_1, σ_{11}) is obtained e.g. by FIML estimation of the complete DSEM.

Note finally from Theorem 4.1 (v) and 4.3 (b) that the standard block-recursive model is sufficient for both (block) predeterminedness and (block) weak exogeneity (again assuming the parameterization satisfies (iii) and (iv)); this may help explain its importance in the development of the theory of simultaneous equations models.

5. SUMMARY AND CONCLUSIONS

Given the pervasive role of the concept of “exogeneity” in econometrics, it is essential to uniquely characterize the implications of claims that certain variables are “exogenous” according to particular definitions. Also, it is useful to have definitions which require minimal conditions and yet are applicable to as wide a class of relevant models as possible. Consequently, general and unambiguous definitions are proposed for *weak*, *strong*, and *super* exogeneity in terms of the joint densities of observable variables and the parameters of interest in given models, thus extending and formalizing the approach in Koopmans [21]. “Exogeneity” assertions are usually intended to allow the analysis of one set of variables without having to specify exactly how a second related set is deter-

¹⁷We are grateful to A. Holly for providing us with this example and, more generally, for pointing out several shortcomings in earlier drafts of this section.

mined and such an analysis could comprise any or all of inference, forecasting, or policy. In each case, the conclusions are conditional on the validity of the relevant “exogeneity” claims (a comment germane to theoretical models also, although we only consider observable variables) and since different conditioning statements are required in these three cases, three distinct, but inter-related, concepts of exogeneity are necessary.

The joint density of the observed variables $x_t = (y_t' z_t')'$, conditional on their past, always can be factorized as the conditional density of y_t given z_t times the marginal density of z_t . If: (a) the parameters λ_1 and λ_2 of these conditional and marginal densities are not subject to cross-restrictions (i.e., there is a cut) and, (b) the parameters of interest (denoted by ψ) can be uniquely determined from the parameters of the conditional model alone (i.e., $\psi = f(\lambda_1)$), then inference concerning ψ from the joint density will be equivalent to that from the conditional density so that the latter may be used without loss of relevant information. Under such conditions, z_t is *weakly* exogenous for ψ , and for purposes of *inference* about ψ , z_t may be treated “as if” it were determined outside the (conditional) model under study, making the analysis simpler and more robust.

Conditions (a) and (b) clearly are not sufficient to treat z_t as if it were fixed in repeated samples, since the definition of weak exogeneity is unspecific about relationships between z_t and y_{t-i} for $i \geq 1$. However, if: (c) y does not Granger cause z , then the data density of $X_t^1 = (x_1, \dots, x_t)'$ factorizes into the conditional density of Y_t^1 given Z_t^1 times the marginal of Z_t^1 and hence $\{z_t\}$ may be treated as if it were fixed. If (a), (b), and (c) are satisfied, then z_t is *strongly exogenous* for ψ and forecasts could be made conditional on fixed future z 's.

Nevertheless, strong exogeneity is insufficient to sustain conditional policy analysis since (a) does not preclude the possibility that while λ_1 and λ_2 are variation free within any *given* “regime,” λ_1 might vary in response to a change in λ_2 between “regimes.” The additional condition that: (d) λ_1 is invariant to changes in λ_2 (or more generally the conditional distribution is invariant to any change in the marginal distribution) is required to sustain conditional *policy* experiments for fixed λ_1 , and z_t is *super exogenous* for ψ if (a), (b), and (d) are satisfied (so that (c) is not necessary either).

In fact, if the generating process of the conditioning variables is susceptible to changes over either sample or forecast periods, then the failure of (d) will invalidate inference and predictions based on the assertion that λ_1 is a constant parameter, whether or not z_t includes “policy variables.” In worlds where policy parameters change, false super-exogeneity assumptions are liable to produce predictive failures in conditional models (see Lucas [23]). Control experiments which involve changes in λ_2 must first establish the super exogeneity of z_t for ψ under the class of interventions considered; we know of no sufficient conditions for establishing such results, but a necessary condition is that the conditional model does not experience predictive failure within sample (see Hendry [18]).

Even in constant parameter worlds (and certainly in worlds of parameter change), the new concepts are distinct from the more familiar notions of predeterminedness and strict exogeneity. Following precise definitions of these

two concepts, it is shown through examples that their formulation in terms of *unobservable* disturbances entails ambiguous implications for inference and that strict exogeneity is neither necessary nor sufficient for inference in conditional models without loss of relevant information. Moreover, models in which predeterminedness is obtained by construction need not have invariant parameters and since predeterminedness is necessary for strict exogeneity, establishing only the latter does not provide a valid basis for conditional prediction or conditional policy. The various concepts are compared and contrasted in detail in closed linear dynamic simultaneous equations systems, and the usefulness of (a) and (b) in clarifying the debate about Wold-causal-orderings is demonstrated.

It is natural to enquire about the testable implications of alternative exogeneity assumptions. Condition (d) is indirectly testable (as noted) via tests for parameter constancy, although as with all test procedures, rejection of the null does not indicate what alternative is relevant and non-rejection may simply reflect low power (so that there are advantages in specifying the regime shift process as in Richard [32]). Condition (c) is common to both strong and strict exogeneity notions and may be testable in the conditional model (see Sims [38] and Geweke [14]) but may also require specification of the marginal density of z_t as in Granger [16]. Also, predeterminedness tests have been the subject of a large literature (see inter alia Wu [44]).

To test weak exogeneity, the conditional and marginal densities could be embedded in a joint density function, although the choice of the latter may or may not generate testable implications. It is somewhat paradoxical to estimate the parameters of a (potentially very complicated) marginal model just to test whether or not one needed to specify that model. Moreover, misspecifications in the marginal model may induce false rejection of the null of weak exogeneity. Nevertheless, Engle [6, 7] considers various weak exogeneity tests based on the Lagrange multiplier principle. Also, on a positive note, while both weak exogeneity and parameter constancy are conjectural features in a conditional modelling exercise, if the data generating process of z_t has changed, but the conditional model has not, then some credibility must attach to the latter since it was hazarded to potential rejection and survived.

Finally, we believe that the new concepts are not only general (being based explicitly on density functions and encompassing worlds of parameter change) and unambiguously characterized (thus clarifying a vital concept in econometrics) but also highlight interesting and novel aspects of familiar problems (as shown in the examples in Section 3).

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